

Spread

Spread describes how much the values vary.

Which synthetic dataset varies more: 4, 5, 6 or 1, 5, 9?

The range

Synthetic data. Range is maximum minus minimum.

What is the range of 3, 4, 4, 7, 12?

A deviation

A deviation is a value minus the mean.

$$x - \bar{x}$$

For mean 5 and value 8, what is the deviation?

Positive and negative deviations

Synthetic values: 2, 4, 6, with mean 4.

What deviations correspond to these values?

Sum of deviations

Synthetic values: 2, 4, 6, with mean 4.

Why would summing their signed deviations fail to measure spread?

Squared deviations

Synthetic values: 2, 4, 6, with mean 4.

What is the sum of squared deviations?

Sample variance

Synthetic data. Sample variance divides the sum of squared deviations by $n - 1$.

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

What is the sample variance of 2, 4, 6?

Sample standard deviation

Standard deviation is the square root of variance.

$$s = \sqrt{s^2}$$

If a sample variance is 4 seconds squared, what is its standard deviation?

Population variance

Synthetic data. Population variance divides by the population size N .

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

What is the population variance when the entire population is 2, 4, 6?

Choose the denominator

We measure 12 randomly selected files from thousands on a server.

Should the sample-variance denominator be 12 or 11?

Standard deviation from frequencies

Synthetic sample: value 2 occurs twice, 4 occurs once, and 6 occurs twice. Use frequencies as weights in the squared deviations.

What sample standard deviation results?

Grouped standard deviation

Synthetic sample: classes $[0,2)$, $[2,4)$, $[4,6)$ have frequencies 2, 1, 2. Replace each observation by its class midpoint.

What estimate of the sample standard deviation results?

Technology and notation

Summary output for a synthetic sample lists $S_x = 2.0$ and $\sigma_x = 1.8$. S_x uses $n - 1$; σ_x uses n .

Which should estimate the population standard deviation from this sample?

Units of spread

File sizes are measured in KB.

What units belong to variance and standard deviation?

Interpret a standard deviation

Synthetic response times have mean 20 seconds and standard deviation 3 seconds.

What does the standard deviation describe?

Compare variability

Two synthetic servers have the same mean response time. Their standard deviations are 2 seconds and 8 seconds.

Which server has more variable response times?

Zero standard deviation

A synthetic dataset has standard deviation zero.

What must be true of every observation?

One value changes

Synthetic values: 4, 5, 6. Replace 6 with 60.

How will the range and standard deviation respond?

Adding a constant

Every synthetic download time increases by 10 seconds.

Does the standard deviation change?

Scaling every value

Convert every response time from seconds to milliseconds by multiplying by 1,000.

How does the standard deviation change?

Distance in standard deviations

A z-score measures distance from a mean in standard-deviation units. Synthetic sizes have mean 40 KB and SD 5 KB.

$$\frac{x - \bar{x}}{s}$$

How many standard deviations above the mean is 55 KB?

Below the mean

Synthetic response times have mean 20 seconds and standard deviation 4 seconds.

What standardized distance corresponds to 12 seconds?

Recovering a data value

Synthetic times have mean 20 seconds and standard deviation 4 seconds.

What value lies 1.5 standard deviations below the mean?

Compare relative positions

Synthetic quiz A: mean 70, SD 5, score 80.

Quiz B: mean 60, SD 10, score 80.

Which score is farther above its mean in standard-deviation units?

Sampling variability of a statistic

Sampling variability is how a statistic changes across repeated samples. Its standard deviation is called its standard error.

Does standard error describe the spread of individual observations or of statistics?

Standard error of the mean

For independent observations with population SD σ , the standard error of the sample mean is $\sigma/\sqrt{n}$.

If $\sigma = 12$ seconds and $n = 36$, what is the standard error?

Chebyshev's bound

For any distribution with finite, positive variance, at least $1 - 1/k^2$ lies within k standard deviations of the mean, $k > 1$.

What minimum fraction lies within two standard deviations?

Empirical rule

For an approximately bell-shaped distribution, about 68%, 95%, and 99.7% lie within one, two, and three standard deviations.

With mean 50 and SD 4, what interval contains about 95%?